Table 9.1: Composition of Air

Gas	Symbol	Content	
Nitrogen	N_2	78.084%	99.99%
Oxygen	O_2	20.947%	
Argon	Ar	0.934%	
Carbon dioxide	CO_2	0.033%	
Neon	Ne	18.20 ppm	
Helium	He	5.20 ppm	
Krypton	Kr	1.10 ppm	
Sulphur dioxide	SO_2	1.00 ppm	
Methane	CH_4	2.00 ppm	
Hydrogen	H_2	0.50 ppm	
Nitrogen oxide	N_2O	0.50 ppm	
Xenon	Xe	0.09 ppm	
Ozone	O_3	0.07 ppm	
Nitrogen dioxide	NO_2	0.02 ppm	
Iodine	I_2	0.01 ppm	

9.2.3 Microflora of Air

Human beings and animals are continuously inhaling the microbes present in the air that cause various infectious diseases. Most of the respiratory tract infections are acquired by inhaling the air pathogen. The microflora of air is studied under two categories such as indoor and outdoor microflora.

Indoor microflora

The air found inside the closed environment (building/ room) is referred as indoor air and the microbes present in this region is called indoor microflora.

Example: *Staphylococcus, Bacillus, Penicillium*.

Outdoor Microflora

The air in the exterior environment is called outdoor air and the microbes that reside there are called outdoor microflora.

Example: *Bacillus, Aspergillus*.

9.2.4 Sources of Microorganisms in Air

Air is not a natural environment for microorganisms as it doesn't contain enough moisture and nutrients to support their growth and reproduction. Soil microorganisms when disturbed by the blow of wind, gets liberated into air and remain suspended there for a long period.

Man-made actions like digging, ploughing the soil may also release soil borne microbes into the air. Microorganisms found in water may also enter air by wind action or tidal action in the form of droplets or aerosols. Air currents may bring the microorganism from plant or animal surfaces into air.

Commensal as well as pathogenic flora of the human beings may enter the air by activities like coughing, sneezing, talking and laughing. The microorganisms are discharged out in different forms of particles which are grouped on the basis of their relative size and moisture content. These are aerosols, droplets, droplet nuclei and infectious dust.

Aerosols

An aerosols are mixture of small liquid or solid particles dispresed in air. Aerosols can be natural or anthropogenic. Example: Dust and smoke.

Droplets

Droplets are formed by sneezing, coughing and talking which consists of saliva and

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